QUANTUM INTERNET

Entangling the world beyond classical communication

ARCHITECTURE • QUANTUM HARDWARE • APPLICATIONS • OPEN CHALLENGES

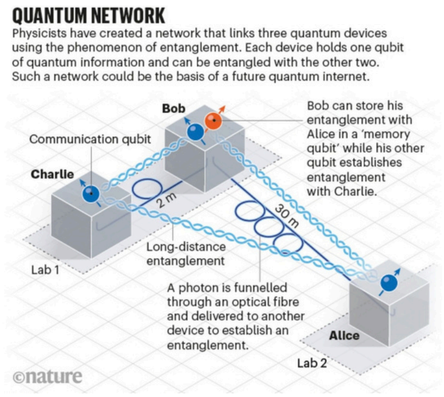
01 WHAT IS A QUANTUM INTERNET?

The Quantum Internet is a network of quantum devices connected by quantum links and supported by classical communication. It enables entanglement distribution, quantum teleportation, and coordination of remote quantum operations—capabilities beyond conventional networking.

WHY IT MATTERS

- Quantum key distribution (QKD) and other quantum-security services.
- Distributed quantum computing and precision clock synchronisation.
- Coordination of remote quantum devices for quantum applications.

FIG. 1 | Entanglement across three nodes

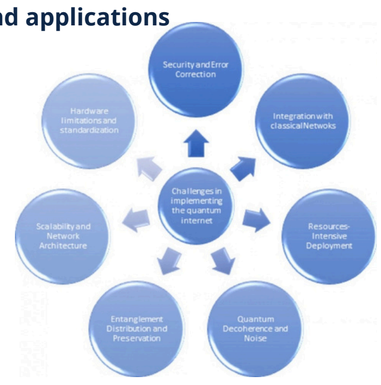


The diagram illustrates how Alice, Bob, and Charlie establish quantum links using photon channels and entanglement. Quantum memories preserve states while the network coordinates link generation and classical communication.

02 THE QUANTUM ADVANTAGE

- **Quantum Entanglement:** Enables shared quantum correlations between distant network nodes.
- **Quantum Teleportation:** Transfers an unknown quantum state using shared entanglement and classical communication.
- **Quantum Security:** Quantum key distribution can detect certain eavesdropping attempts under specified security assumptions.
- **Distributed Quantum Computing:** Enables research into quantum operations across interconnected quantum processors.

FIG. 2 | Stages and applications



Quantum networks progress from key distribution and trusted nodes toward end-to-end entanglement, quantum memories, error correction, and distributed quantum computing. These capabilities depend on advances in hardware and network protocols.

03 HOW QUANTUM COMMUNICATION WORKS

- **CREATE ENTANGLEMENT :** neighbouring nodes generate and distribute an entangled pair (often called a Bell pair).
- **EXTEND THE LINK :** an intermediate node performs a joint measurement to carry entanglement across adjacent links; this is entanglement swapping.
- **TELEPORT THE STATE :** the sender measures the input qubit together with its entangled qubit and sends the measurement result over a classical channel. The receiver uses that result to reconstruct the state.
- **COORDINATE :** classical messages schedule operations and communicate outcomes. Teleportation consumes the shared entanglement and does not enable faster-than-light communication.

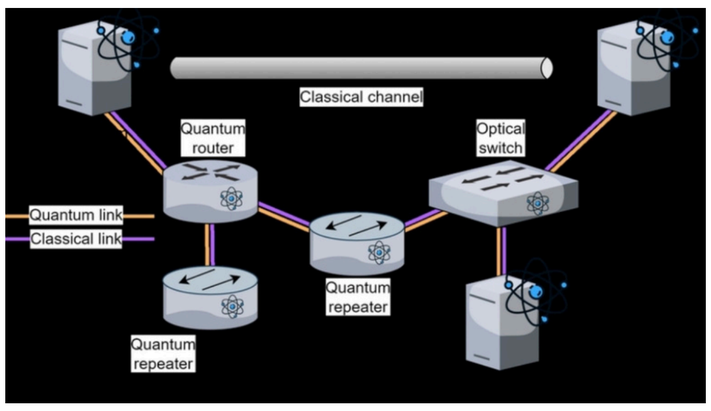
KEY TAKEAWAY

The Quantum Internet is a complementary networking paradigm that could enable entanglement-based communication, distributed quantum computing, and networked quantum sensing. Its practical development depends on reliable entanglement distribution, quantum memory, interoperable hardware, and scalable network protocols.

WHY QUANTUM REPEATERS ARE DIFFERENT

A classical repeater regenerates or amplifies a signal. A quantum repeater cannot simply copy an unknown qubit. Instead, it helps establish entanglement over neighbouring shorter links and uses entanglement swapping to extend it. Purification or quantum error correction may improve fidelity, but both require additional quantum resources and operations.

FIG. 3 | A hybrid quantum network

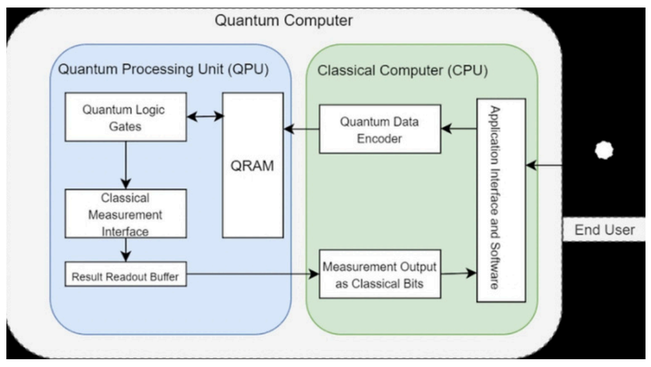


Quantum links distribute quantum states and entanglement, while classical links carry control instructions and measurement outcomes. Routers, switches, and repeaters coordinate network operations.

04 CORE BUILDING BLOCKS

- **QUANTUM SOURCES:** produce single photons or entangled photon pairs; useful sources should be reliable and produce states suitable for the intended protocol.
- **PHOTONIC CHANNELS:** optical fibre and free-spacelinks carry quantum states, but loss increases with distance. Satellite links may help bridge long distances.
- **DETECTORS:** register arriving photons and measurement outcomes; efficiency and noise affect link performance.
- **QUANTUM MEMORIES:** store quantum states long enough to coordinate links that succeed at different times. Memory lifetime and read/write quality are key constraints.
- **QUANTUM REPEATERS:** combine shorter entanglement links through swapping; purification or quantum error correction can improve reliability at additional resource cost.
- **CLASSICAL CONTROL & NETWORK SOFTWARE:** manage timing, link requests, routing and measurement messages. Hardware and protocol design must evolve together.

FIG. 4 | Quantum computer and classical control



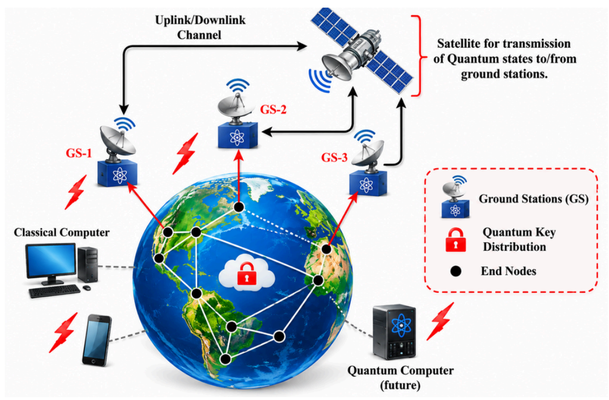
The QPU processes quantum states, while the classical controller prepares tasks, coordinates operations, and interprets measurement results. Network links and entanglement can connect quantum processors.

05 FROM LABORATORY LINKS TO A NETWORK

Quantum networking requires more than physical connections. Nodes must establish entanglement, coordinate operations, and manage communication resources.

- **Experimental progress:** Quantum network protocols have been demonstrated over a 14.4 km urban optical-fiber link.
- **Network coordination:** Entanglement swapping can connect shorter links to establish entanglement between more distant nodes.
- **Scaling challenge:** Photon loss, probabilistic entanglement generation, and limited quantum memory complicate network expansion.

FIG. 5 | Satellite-assisted quantum links



Satellite-based quantum links could connect distant ground stations beyond the reach of practical direct fibre links. Performance depends on photon sources, pointing and tracking, atmospheric conditions, and detector efficiency.

ABOUT THIS POSTER

This poster synthesises the four supplied research PDFs: QRNA architecture, quantum hardware, networking challenges and the 2025 building-blocks review. All six diagrams are the images supplied by the requester; no replacement or AI-generated images were added.

06 POTENTIAL APPLICATIONS

SECURE COMMUNICATION

QKD enables two parties to establish shared secret keys and detect certain eavesdropping attempts under specified security assumptions.

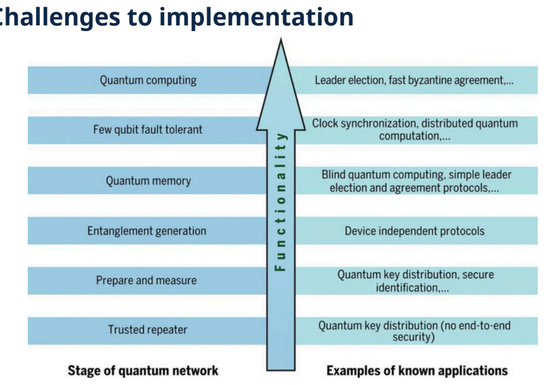
DISTRIBUTED QUANTUM COMPUTING

Entanglement and teleportation could connect remote quantum processors, subject to hardware and network limitations.

SENSING & SYNCHRONISATION

Networked quantum systems could support precision sensing, interferometry, and clock synchronisation.

FIG. 6 | Challenges to implementation



Quantum networking faces challenges including decoherence, photon loss, entanglement degradation, hardware limitations, error management, interoperability, and scalability. These constraints are interconnected.

07 THE HARDEST PROBLEMS

- **PHOTON LOSS & DECOHERENCE:** Photon loss reduces transmission success, while environmental interactions degrade quantum-state coherence.
- **ENTANGLEMENT SCALABILITY:** Long-distance entanglement requires reliable link coordination, sufficient resources, and entanglement swapping.
- **QUANTUM MEMORY:** Limited storage lifetime and imperfect retrieval complicate coordination between network operations.
- **NETWORK INTEROPERABILITY:** Different quantum devices require compatible interfaces, protocols, and classical control.

08 RESEARCH OUTLOOK

Research is advancing toward scalable, interconnected quantum networks.

KEY RESEARCH DIRECTIONS

- **Quantum Repeaters:** Improve long-distance entanglement distribution and reduce the impact of photon loss.
- **Quantum Memories:** Develop reliable memories with longer storage times and higher fidelity.
- **Quantum Network Protocols:** Improve entanglement routing, resource allocation, and network coordination.
- **Scalability & Interoperability:** Enable reliable connections between heterogeneous quantum devices.

FUTURE GOAL: Enable practical quantum communication, distributed quantum computing, and networked quantum sensing.

09 CLASSICAL VS QUANTUM NETWORKING

CLASSICAL INTERNET

Classical networks transmit information as bits using electrical, optical, or wireless signals. Data can generally be copied, stored, routed, and retransmitted. Security relies on cryptographic protocols and their implementations.

QUANTUM INTERNET

Quantum networks distribute quantum states and entanglement between nodes. Unknown quantum states cannot be perfectly cloned. Quantum teleportation uses shared entanglement and classical communication, while practical networks must manage photon loss, decoherence, and limited quantum resources.

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